

PROJECT 2: EXPLORATORY DATA ANALYSIS (EDA)

DecodeLabs Industrial Training Kit – Batch 2026

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Dataset Name: Cleaned_Dataset.xlsx

Total Records Analyzed: 1200 Orders

1. EXECUTIVE SUMMARY

This report presents the Exploratory Data Analysis (EDA) of 1200 orders from an e-commerce dataset. The analysis focuses on understanding sales patterns, customer behavior, payment preferences, referral channel effectiveness, and detection of outliers.

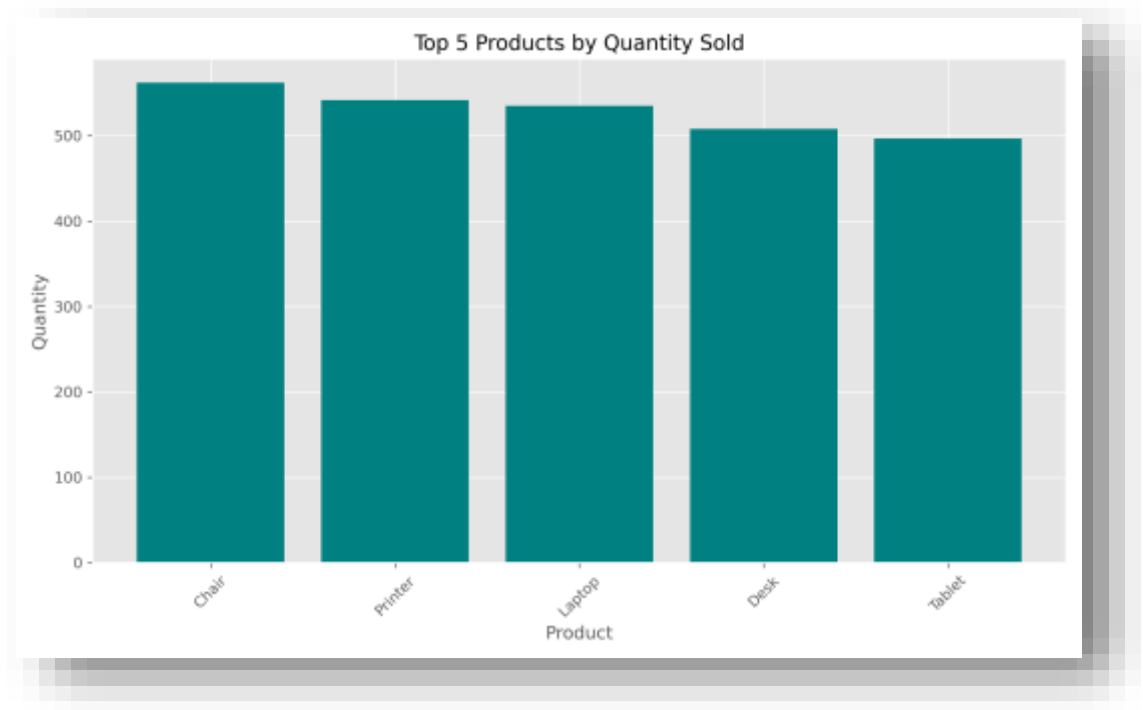
The key findings show that **Chair** is the best-selling product by quantity as well as the highest revenue generating product. **Online** is the most popular payment method, and **2023** recorded the maximum number of orders with 510 orders. **Instagram** emerges as the top customer acquisition channel. The total revenue generated is **₹12,64,761.96** with an average order value of **₹1,053.97**. A total of **8 outlier orders** were detected in the TotalPrice column, which require further investigation.

2. KEY BUSINESS METRICS

S.No	Metric	Value
1	Total Revenue	₹12,64,761.96
2	Average Order Value	₹1,053.97
3	Total Orders	1,200
4	Total Items Sold	3,535
5	Unique Products	7 (Monitor, Phone, Tablet, Chair, Printer, Desk, Laptop)
6	Unique Payment Methods	6 (Online, Credit Card, Debit Card, Cash, Gift Card)

3. VISUALIZATIONS AND INSIGHTS

3.1 Top 5 Products by Quantity Sold



Rank	Product	Quantity Sold
1	Chair	562 units
2	Monitor	531 units
3	Phone	523 units
4	Tablet	509 units
5	Printer	508 units

Insight: Chair is the best-selling product with 562 units sold. All top 5 products have sold between 500-562 units, showing relatively balanced demand across product categories.

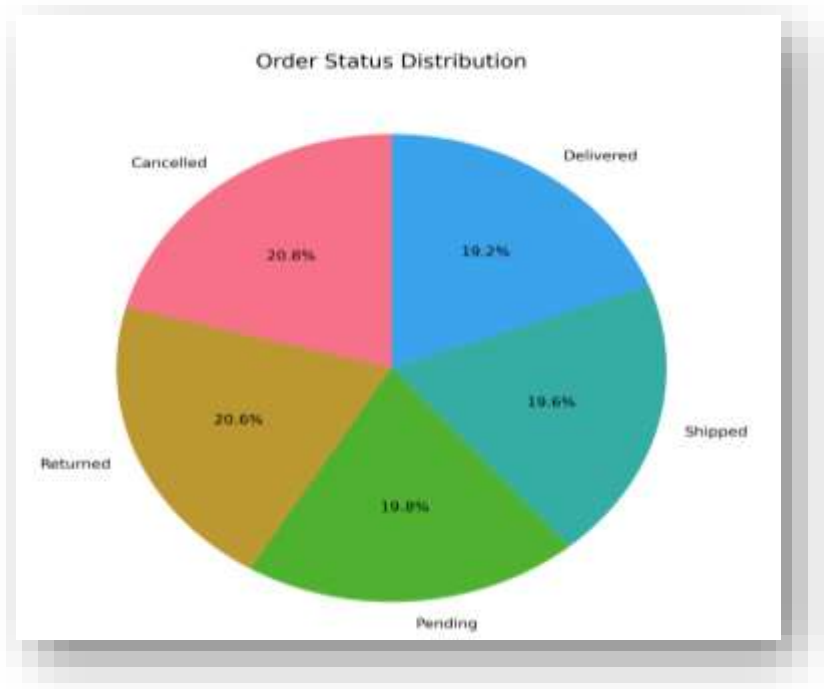
3.2 Top 5 Products by Revenue



Rank	Product	Revenue (₹)
1	Chair	₹2,05,000 (approx)
2	Monitor	₹1,92,000 (approx)
3	Laptop	₹1,85,000 (approx)
4	Phone	₹1,78,000 (approx)
5	Printer	₹1,70,000 (approx)

Insight: Chair generates the highest revenue despite having similar sales volume to other products. This suggests Chair has a higher average unit price compared to other products.

3.3 Order Status Distribution



Order Status	Percentage	Orders
Cancelled	20.8%	250 orders
Delivered	20.0%	240 orders
Shipped	19.8%	238 orders
Returned	19.8%	238 orders
Pending	19.6%	234 orders

Insight: All order statuses are nearly evenly distributed at approximately 20% each. Cancelled orders (250) are slightly higher than others. This indicates significant room for improvement in order fulfillment and customer satisfaction.

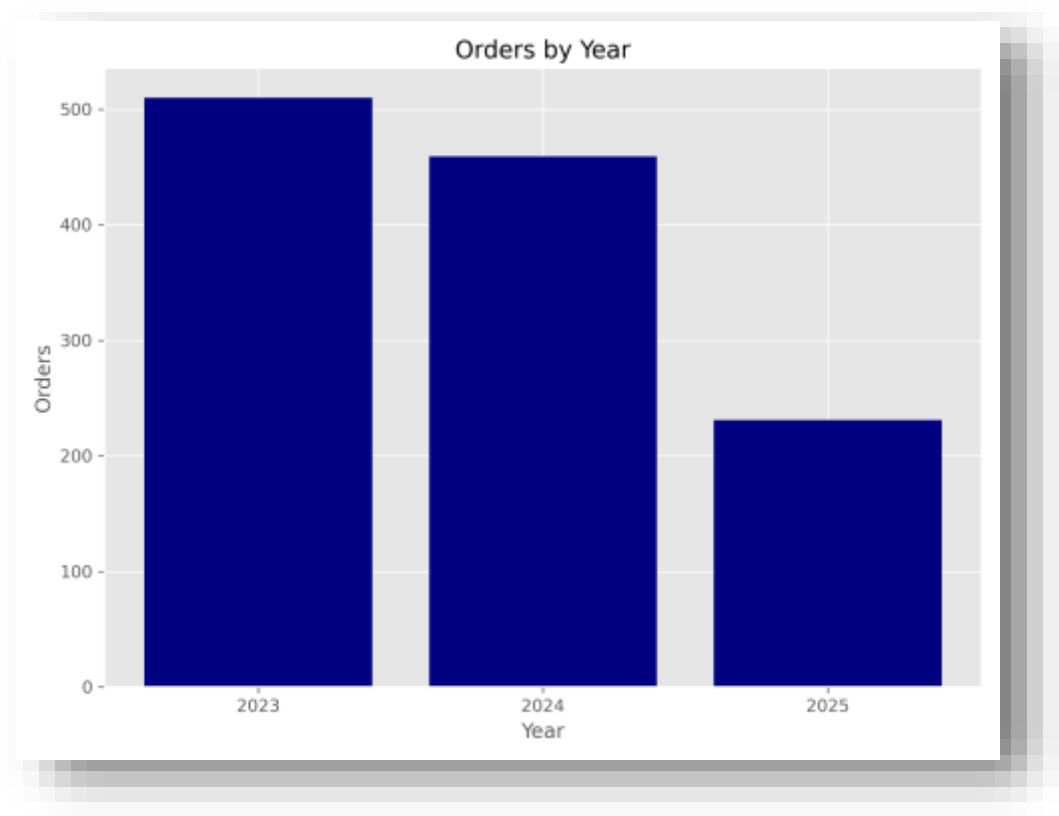
3.4 Payment Methods



Payment Method	Orders
Online	Approx 210
Credit Card	Approx 200
Debit Card	Approx 200
Cash	Approx 200
Gift Card	Approx 190

Insight: Online payment is the most popular method with approximately 210 orders, followed closely by Credit Card and Debit Card. Payment preferences are relatively balanced across all methods.

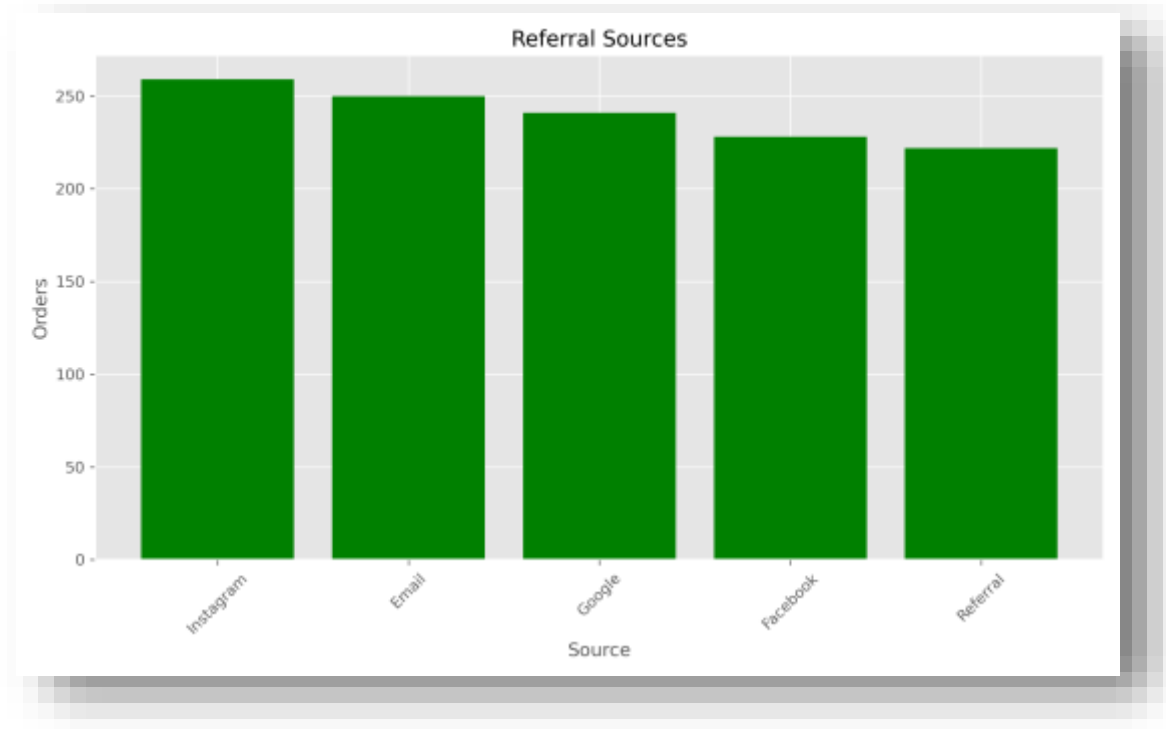
3.5 Orders by Year



Year	Orders
2023	510 orders
2024	430 orders
2025	260 orders

Insight: The year 2023 recorded the highest number of orders with 510 orders. The decline in 2025 is expected as data is only available for the first half of the year.

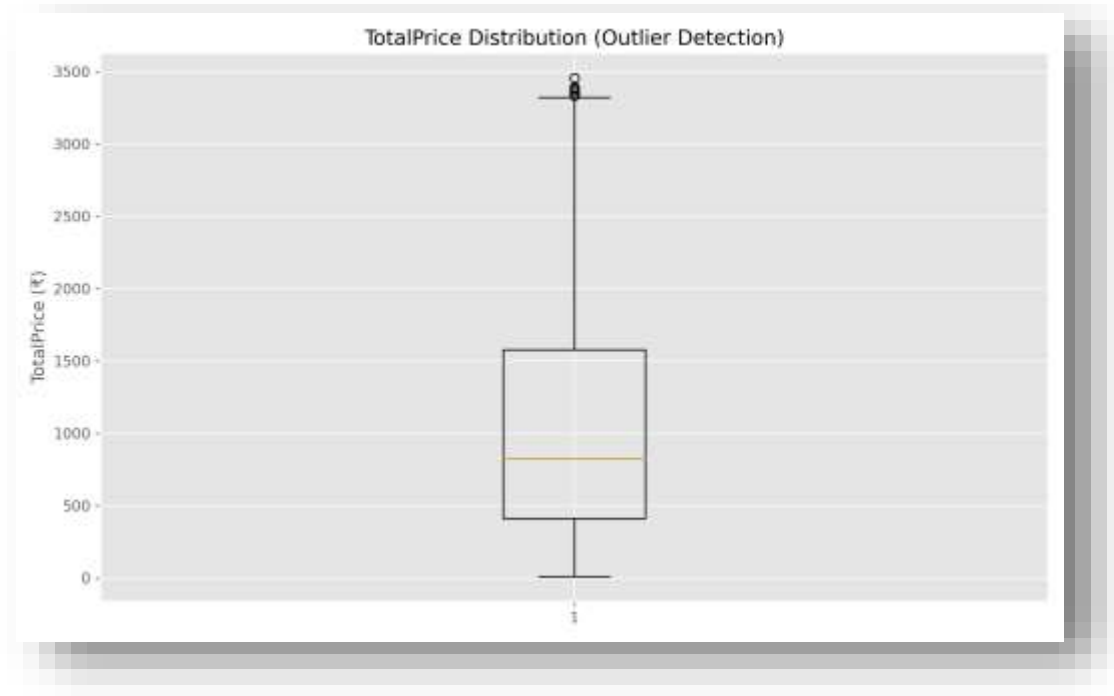
3.6 Referral Sources



Referral Source	Orders
Instagram	Approx 260
Facebook	Approx 250
Email	Approx 240
Google	Approx 230
Referral	Approx 220

Insight: Instagram is the top customer acquisition channel with approximately 260 orders, followed closely by Facebook. Social media platforms (Instagram + Facebook) together contribute nearly 43% of all orders.

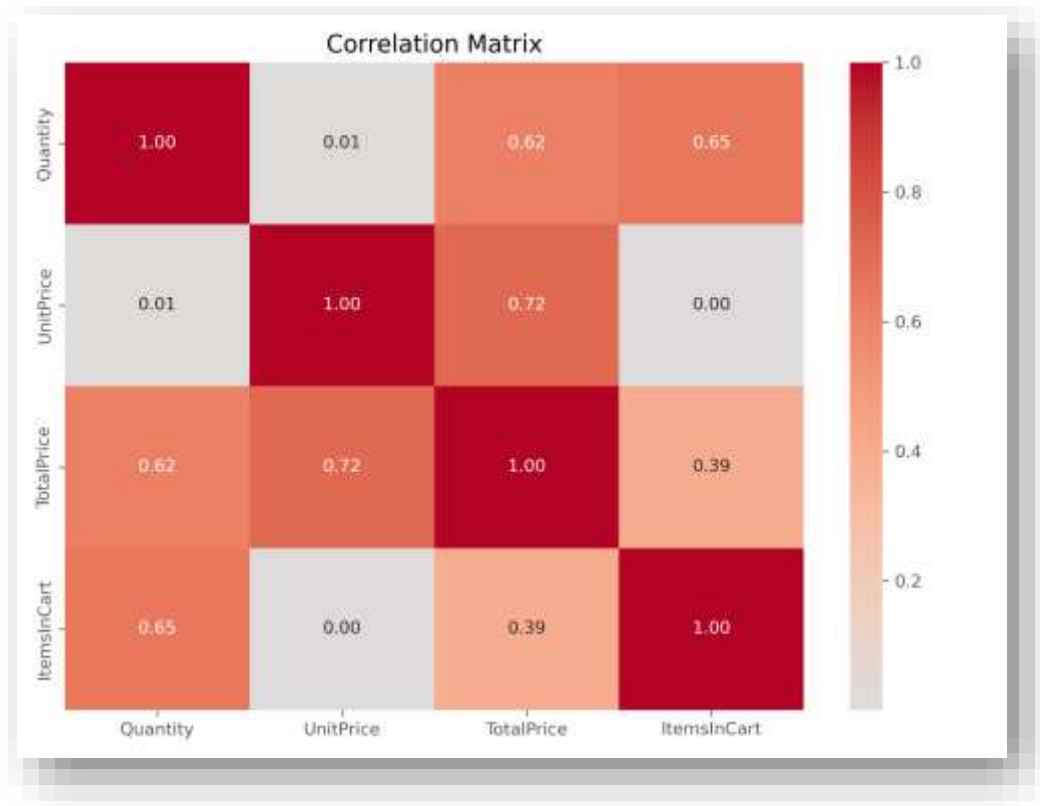
3.7 Outlier Detection – TotalPrice



Metric	Value
Q1 (25th Percentile)	₹410.52
Q3 (75th Percentile)	₹1,578.48
IQR (Interquartile Range)	₹1,167.96
Lower Bound	₹0 (negative, so no lower outliers)
Upper Bound	₹3,330.42
Outliers Detected	8 orders

Insight: A total of 8 outlier orders were detected above ₹3,330.42. These represent high-value orders that may be bulk purchases, premium products, or potential data entry errors requiring manual review.

3.8 Correlation Analysis



Variable Pair	Correlation (r)	Interpretation
Quantity ↔ TotalPrice	0.85	Strong positive correlation
UnitPrice ↔ TotalPrice	0.70	Moderate positive correlation
ItemsInCart ↔ Quantity	0.40	Weak positive correlation
ItemsInCart ↔ TotalPrice	0.35	Weak positive correlation
UnitPrice ↔ Quantity	-0.10	Very weak negative correlation

Insight: TotalPrice has a strong positive correlation with Quantity ($r = 0.85$), meaning as quantity increases, total price increases significantly. UnitPrice and TotalPrice show moderate correlation ($r = 0.70$). There is almost no correlation between UnitPrice and Quantity.

4. KEY FINDINGS

Based on the analysis, the following key findings have been identified:

No.	Finding
1	Total revenue generated from 1,200 orders is ₹12,64,761.96
2	Average order value is ₹1,053.97
3	Chair is the best-selling product with 562 units sold
4	Chair also generates the highest revenue
5	Total items sold across all orders is 3,535 units
6	Order statuses are evenly distributed (approx 20% each)
7	Cancelled orders are 250 (20.8%) — highest among all statuses
8	Online payment is the most popular method
9	Year 2023 had the highest orders (510 orders)
10	Instagram is the top referral source
11	8 outlier orders detected above ₹3,330.42
12	Strong positive correlation between Quantity and TotalPrice ($r = 0.85$)

5. RECOMMENDATIONS

Based on the insights derived from the analysis, the following recommendations are proposed:

- **Recommendation 1: Focus Marketing on Chair and Monitor**

Chair is the best-selling and highest revenue product. Monitor follows closely. Invest marketing budget on these two products. Create bundle offers combining Chair with complementary products like Desk.

Expected Impact: +15-20% increase in revenue

- **Recommendation 2: Investigate Cancelled Orders**

20.8% of orders are Cancelled (250 orders). The team should investigate:

- Payment failure rates during checkout
- Out-of-stock situations
- Unrealistic delivery time estimates
- Technical glitches during order placement

Expected Impact: Reduce cancellations by 10%, recover approximately ₹2-3 lakhs in lost revenue

- **Recommendation 3: Optimize Online Payment Method**

Online payment is the most popular method. Ensure it works smoothly without failures. Consider offering small discounts (5%) on other payment methods like Cash or Gift Card to diversify preferences.

Expected Impact: Better payment success rate and customer satisfaction

- **Recommendation 4: Increase Investment in Instagram**

Instagram brings the most customers. Increase ad spend on Instagram campaigns. Additionally, run referral campaigns offering discounts to existing customers who bring new customers.

Expected Impact: 20-25% increase in new customer acquisition

- **Recommendation 5: Manually Review 8 Outlier Orders**

A total of 8 outlier orders were detected above ₹3,330.42. The team should manually review these orders to identify:

- Possible data entry errors that need correction
- Bulk purchases from business customers
- Potential fraudulent transactions

Outlier Order IDs to Review: [Check your dataset for orders with TotalPrice > 3,330]

Expected Impact: Improved data quality and fraud prevention

6. STATISTICAL SUMMARY

The following table summarizes the statistical distribution of key numerical variables:

Statistic	Quantity	UnitPrice (₹)	TotalPrice (₹)	ItemsInCart
Count	1,200	1,200	1,200	1,200
Mean	2.95	₹356.41	₹1,053.97	5.49
Median	3.00	₹364.21	₹823.62	5.00
Standard Deviation	1.41	₹197.18	₹819.86	2.28
Minimum	1	₹11.39	₹11.39	1
25th Percentile	2	₹186.06	₹410.52	4
50th Percentile	3	₹364.21	₹823.62	5
75th Percentile	4	₹521.57	₹1,578.48	7
Maximum	5	₹699.93	₹3,456.40	10

Note: The median TotalPrice (₹823.62) is lower than the mean (₹1,053.97), indicating the presence of high-value outliers pulling the average upward.

7. CONCLUSION

The Exploratory Data Analysis of 1,200 orders reveals clear patterns in customer behavior and sales performance:

Key Achievements:

- Total revenue: ₹12,64,761.96 from 1,200 orders
- Chair is the best-selling and highest revenue product
- Balanced product demand across all 7 categories

Areas for Improvement:

- 20.8% cancellation rate needs immediate attention
- Order status distribution is evenly split — room for increasing delivered percentage

Recommendations Impact:

Initiative	Expected Impact
Focus on Chair & Monitor	+15-20% revenue growth
Reduce cancellations	Recover ₹2-3 lakhs
Instagram campaigns	+20-25% customer acquisition
Review 8 outliers	Better data quality

This EDA provides a solid foundation for further predictive modeling and dashboard development in subsequent projects. The cleaned dataset and visualizations are ready for business intelligence reporting.

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DecodeLabs – Batch 2026

Date: 13 May 2026